

1 **Key-words:** Apidae, Amplicon sequencing, Interaction, Metabarcoding, Network, Pollinator, Plant-
2 pollinator interaction, Gut microbiome

3 **Abstract**

4 Our understanding of plant-pollinator interaction networks hinges on the methods used to describe their
5 nodes and links. Most networks are built from field observations that may overlook many consumer-
6 resource links, and these networks lack descriptive links that characterize interaction types and out-
7 comes. Towards a more complete approach for building interaction networks, we compare plant inter-
8 actions from the wild pollinator species, *Bombus pascuorum*, recorded by three methodologies with
9 different implications for interaction outcomes. We compare floral visitation interactions obtained
10 from field observations, plant consumption interactions revealed by metabarcoding of gut contents,
11 and pollen transport interactions detected by metabarcoding of corbicular pollen loads. Our approach
12 adds functional context to plant-pollinator network links and reveals new interactions. We show that
13 both metabarcoding approaches increase the number of interactions and reveal links that were over-
14 looked by field observations of visitation, highlighting plant taxa that are not pollinator-dependent, yet
15 constitute important dietary resources. Paired with floral diversity surveys, gut-content results also
16 reveal seasonal patterns in the spatial extent and functional diversity included in forage, which other
17 methodologies fail to demonstrate. Metabarcoding data analyzed at the individual specimen level fur-
18 ther contribute heterogeneity in plant resource use between pollen transport and consumption. Metabar-
19 coding methodologies capture greater spatial, temporal, and taxonomic ranges, while field observations
20 provide validating datasets with higher taxonomic precision. Our results show that integrating visita-
21 tion, transport, and consumption data changes network topology and the roles of plant nodes, offering

22 a more nuanced and complete map of interactions with clearer priorities for management. We advocate
23 for defining links explicitly by their functions and combining methods to account for hidden structure
24 in ecological networks.

25 **Data and Code for peer-review**

26 All data and code used for the analyses in this manuscript are provided on an Anonymous GitHub repos-
27 itory (https://anonymous.4open.science/r/B_pascuorum_interaction_networks). All raw amplicon se-
28 quencing data will be deposited in a project in the European Nucleotide Archive upon final manuscript
29 acceptance.

30 **1. Introduction**

31 Pollination is a critical ecosystem service that is currently threatened by global anthropogenic change,
32 including habitat loss, intensifying agriculture, pathogens, and invasive species (Klein et al., 2006).
33 Pollinators crucially support the reproduction of 94% of wild flowering plants and 75% of crop species
34 (Vanbergen & Insect Pollinators Initiative, 2013), contributing to 35% of global food production (Klein
35 et al., 2006). Despite the clear importance of understanding plant-pollinator interactions, our knowledge
36 of interaction diversity remains incomplete, as the methodological approach to studying plant-pollinator
37 interactions has historically been biased towards the plants (Bosch et al., 2009; Evans & Kitson, 2020).
38 As a consequence, the well-established relationship between pollinator diversity and the productivity of
39 plant communities (Artamendi et al., 2025; Katumo et al., 2022; Woodcock et al., 2019) lacks an equally
40 developed mirrored perspective, describing the floral diversity that supports pollinator populations.

41 Network theory provides a useful framework to summarize patterns of plant–pollinator interaction

(Burkle & Alarcón, 2011), but most studies under this framework have yet to account for how the scope of networks is influenced by the interaction types that define links. Existing methodologies for reconstructing interaction networks tend to emphasize structural patterns, while overlooking the functional outcomes of interactions that are critical for understanding how plant communities support pollinators (Quintero et al., 2022). In *Apidae* species, for example, plant interactions may have several outcomes. Bees consume plant material, including pollen, nectar, or even plant tissue (Pashalidou et al., 2020; Vaudo, 2015). They also collect pollen on their corbicula for transport to the nest for feeding drones and larvae (Leach & Drummond, 2018; Vaudo, 2015). Finally, visitation of the reproductive parts of flowers can have various outcomes for both the plant and pollinator, including pollen transfer (Emer & Memmott, 2023) and pathogen transfer (Lignon et al., 2024). Interaction networks generally represent only one of these outcomes, although each is important to understanding how plant taxa support pollinators.

The importance of different outcomes in plant-pollinator interactions becomes clear when considering the biodiversity necessary to support pollinators across life stages. Because the resources needed for foraging adult pollinators are different from those needed by larvae or other colony members (Leach & Drummond, 2018; Vaudo, 2015), transported pollen may not completely represent the interactions necessary to sustain adult pollinators. This is especially true for bumblebees (*Bombus spp.*), which are able to evaluate pollen resource quality to discern foraging choices (Leonhardt & Blüthgen, 2012; Timberlake, de Vere, et al., 2024). Bumblebees make trial-and-error floral visits in order to find adequate forage (Selva et al., 2024), which may result in pollen transport without consumption. Conversely, consumption, or simply visitation, may occur without resulting in transport (Popic et al., 2012). Ac-

63 counting for different interaction outcomes, such as visitation, transport, and consumption, is a critical
64 next step in representing the network of plant diversity used by pollinators.

65 Incorporating the pollinator-perspective by leveraging complimentary methodologies can produce a
66 comprehensive understanding of the contexts of plant-pollinator interactions. Microscopy and DNA
67 analyses of pollen load samples sourced from insect specimens often identify greater plant species
68 diversity within interaction networks compared to only field observations of floral visitation (Baksay
69 et al., 2022; Bosch et al., 2009). Additionally, studies adopting a pollinator-centric view have revealed
70 greater detail in forage preference trends, such as how pollinators use quality or quantity-based forage
71 strategies (Selva et al., 2024; Timberlake, de Vere, et al., 2024), seasonal changes (Leponiemi et al.,
72 2023), life cycle timing, and metabolic specialization (Vaudo, 2015)

73 Genetic tools can detect plant-pollinator interactions that may be unobserved in pollen microscopy
74 and field surveys (Arstingstall et al., 2021; Bell et al., 2016; Lowe et al., 2022; Pornon et al., 2017),
75 and target specific interaction types. Metabarcoding of conserved gene regions from pollen samples
76 can complement field observations of visitation (Arstingstall et al., 2021; Bell et al., 2017), increasing
77 species detection by 9 - 144% (Baksay et al., 2022; Milla et al., 2022; Smart et al., 2017) and network
78 sampling completeness up to 30%, while reducing exaggeration of specialization (Arstingstall et al.,
79 2021) and revealing interactions beyond the traditionally surveyed floral community (de Vere et al.,
80 2017; Milla et al., 2022). Advances in the reliability and accessibility of sequencing technologies have
81 made these approaches more feasible for studying plant-pollinator interactions.

82 Most pollinator interaction network studies that apply metabarcoding focus on the external pollen loads
83 of bees or pollen stored in nest reserves of honey and beebread (Baksay et al., 2022; Devriese et al.,

84 2024; Leontidou et al., 2021; Leponiemi et al., 2023; Selva et al., 2024), despite limitations of these
85 sampling targets. Pollen in these samples can come from the environment, even including accumulation
86 of windborne material (Negri et al., 2015). To account for this, past studies have ignored detections
87 of wind-pollinated taxa (Pornon et al., 2017; Tanaka et al., 2020), potentially biasing results, given
88 that many plant taxa are both wind- and insect-pollinated (Saunders, 2018). A more fundamental issue
89 with externally carried pollen and nest reserves is their restricted ability to represent interaction types.
90 Corbicular pollen provides an easily obtained sample, containing pollen from one or more plant species
91 collected for transport to the nest for brood feeding (Leach & Drummond, 2018; Vaudo, 2015), only
92 directly representing pollen transport interactions (Arstingstall et al., 2021). This pollen is often the
93 central sample type used to represent foraging networks, where it is considered synonymous with diet
94 and successful pollination interactions (Shi et al., 2025), likely overstating the function of corbicular
95 pollen.

96 Pollinator intestinal tracts (hereafter: guts) represent a source for observing consumption of pollen and
97 other plant material (Haag et al., 2023; Li et al., 2025; Mayr et al., 2021). Plant DNA detected in gut
98 contents can reveal interactions with consumption as the exclusive outcome, which can include nec-
99 tar robbing (Popic et al., 2012) and occasional herbivory (Pashalidou et al., 2020). The gut-content
100 approach can also account for environmental contamination in external pollen and nest stores by high-
101 lighting oversights resulting from the exclusion of interactions with the anemophilous and partially-
102 anemophilous plant taxa in external pollen studies. There is an accumulating body of evidence sup-
103 porting the idea that pollinators regularly search across functional groups of the plant community to
104 meet their nutritional needs (de Vere et al., 2017; Ibiyemi et al., 2025; Milla et al., 2022; Pojar, 1973;

105 Selva et al., 2024; Tanaka et al., 2020; Terrell & Batra, 1984; Timberlake, de Vere, et al., 2024; Wood
106 et al., 2022), although little attention has been given to these observations as a potentially important
107 part of plant-pollinator networks (Saunders, 2018). This understudied component of pollinator forag-
108 ing together with the surprising lack of genetic analyses of pollinator gut contents, represents a clear
109 knowledge gap and an opportunity to uncover finer detail in plant-pollinator interaction networks.

110 We assess how metabarcoding of pollinator gut contents can complement or challenge the character-
111 ization of plant-pollinator interaction networks described by more common methodologies, including
112 field surveys of plant-pollinator interactions, and external pollen load metabarcoding. To this end, we
113 compare interaction networks constructed from each of these methodologies for a single model polli-
114 nator, *Bombus pascuorum*, an easily identified bumblebee common to most of Europe (Lecocq et al.,
115 2015). Our focus on a single pollinator species holds pollinator identity constant and attributes dif-
116 ferences in network structure to methodology, rather than to variation among pollinator species. We
117 hypothesize that the consumption interactions detected in gut metabarcoding will include a network of
118 plant taxa distinct from those detected by other methodologies. Although we expect overlap between
119 networks constructed by different methodologies, we expect to observe previously overlooked inter-
120 action network structure, including new links and significance of network links. Ideally, the resulting
121 combination of observations will generate a network that will elevate our capacity to detect meaning-
122 ful plant-pollinator interactions, and learn more about interaction types and implications for pollinator
123 health.

124 2. Methods

125 Our sample collection was conducted in Gorbeia Natural Park (coord: 43.068, -2.796) , a protected

126 area in northern Spain, under permission from local authorities (Bizkaiko foru aldundia, Reference:
127 2023AM36). Within Gorbeia, we selected 16 sampling sites located within the mixed zones of meadows
128 and shrublands found at higher elevations. We conducted fieldwork from early April to the end of July,
129 2023 covering the main flowering period and peak annual pollinator activity in Gorbeia. On each
130 sampling day during this timeframe, we visited field sites in pairs. Sampling days were organized into
131 six periods, in which we sampled each site pair once per period. We conducted three types of surveys
132 during daily peaks of pollinator activity, including floral diversity surveys (“flower counts”), interaction
133 transect surveys, and *Bombus pascuorum* specimen collection for amplicon sequencing analyses.

134 *Interaction transects and floral resource availability surveys*

135 We used one 250 m transect at each site for both interaction transect and flower count surveys, recording
136 observations within a ~2 m wide transect line. Interaction surveys were conducted three times per day,
137 each lasting 1 h. All insects observed contacting the reproductive parts of herbaceous flowers within the
138 transect were recorded; for this study, we retained only *Bombus pascuorum* interaction data. Surveys
139 were spaced by ~2 hours (~11:00, ~13:00, ~15:00), and transects were walked at a constant pace to cover
140 the full length within an hour. For each site and sampling period, one flower count was conducted by
141 recording all of the flowering herbaceous species within the transect.

142 *Pollinator specimens*

143 For every period visit at each site, we collected up to five *B. pascuorum* specimens for molecular
144 analyses (N = 126). We brought specimens back from the field and froze them at -20°C until processed.
145 In the lab, we extracted the entire gut and honey stomach of *B. pascuorum* individuals. Additionally,

146 if present, we collected pollen from the corbicula of specimens into sterile 1.5 mL centrifuge tubes.
147 Pollen samples were stored individually by specimen sample at -20°C.

148 *Gut-content DNA extraction*

149 Genomic DNA was extracted from *B. pascuorum* guts using the NucleoSpin® 96 Soil kit (Macherey-
150 Nagel, Düren, Germany). We followed the kit manufacturer protocol, only adjusting centrifuge spin
151 duration to account for differing maximum velocity available within our centrifuge (See Supporting
152 Information). Nanodrop spectrophotometry was used to quantify DNA concentration and purity of the
153 extracts by measuring reflectance at 260/230 nm wavelengths. To avoid site and period bias, all samples
154 were randomized before the DNA extraction.

155 *DNA extraction from corbicular pollen pellets*

156 DNA was extracted from pollen pellets (N = 25) using the Machery-Nagel NucleoSpin® 8 Food kit,
157 including additional initial steps recommended by the kit's supplementary protocol for pollen DNA ex-
158 traction (See Supporting Information). The Qubit high sensitivity dsDNA kit (Thermo Fisher Scientific)
159 was used to quantify DNA extract concentrations for randomly selected samples.

160 *Amplicon sequencing*

161 For gut-content samples, DNA was amplified in duplicate using the DFD forward and ASDFAS reverse
162 primers. We used a dual-indexed amplicon multiplexing approach to generate our metabarcoding li-
163 brary, as described previously by Donald et al. (2022). Briefly, we performed a first step amplification
164 using the modified (See supporting information) internal transcribed spacer 2 primer pair, ITS-S2F (AT-
165 GCGATACTTGGTGTGAAT) (Chen et al., 2010) and ITS4R (TCCTCCGCTTATTGATATGC) (White

et al., 1990), in duplicate. The final library was sequenced on an illumina MiSeq 3000 instrument for 300 cycles in paired-end mode at the Ecological Genetics laboratory, Bioeconomy Science Institute, Auckland, New Zealand (Detailed methods provided in Supporting Information). Pollen DNA samples were sequenced and processed by NOVOGENE (Planegg, Germany) using the same ITS2 primer pair on an illumina platform.

Bioinformatics: taxonomic assignment and contaminant analysis

For all samples, raw reads were processed using the DADA2 bioinformatics pipeline (Callahan et al., 2016), producing two key datasets: a sample x ASV counts matrix and reference ASV fasta file. Reference ASVs were used to call taxonomic identities using DADA2 in conjunction with the published ITS2 reference database of Bell (2021). Taxonomy data were combined with the sample x ASV matrix and sample metadata using phyloseq (McMurdie & Holmes, 2013) and downstream quality control and statistical analyses were performed in the R analysis environment. We used iNEXT to quantify sampling completeness for our data using sample coverage (Hsieh et al., 2025). Coverage summaries were calculated at the observed sample size only, excluding extrapolated values. For contaminant and misidentified ASV removal, we used a three-step screening process. First, ASVs were analyzed for contaminants using the R package, decontam (Davis et al., 2018). Second, we conducted a BLAST search using ITS2 Database (Ankenbrand et al., 2015) to verify taxa that were identified by singleton ASVs within our results. Finally, the remaining list of taxa was screened against a locally specific herbarium (Agut & Hermosilla, 2025).

Statistical analysis

We compared the results of each methodology, including comparison across time, and individual spec-

imens. As an initial broad test of whether the methodologies detected interactions with different plant communities, we used binary presence-absence matrices to compare the communities detected by each methodology for each sampling day. Data were aggregated by sampling day for all sets of observations. Community composition was contrasted using the Raup-Crick dissimilarity index in a PERMANOVA test within the R package, *vegan* (Oksanen et al., 2024) with methodology as the independent variable. Further pairwise comparisons of these data were made by subsetting the dissimilarity matrix used in the first test by each unique methodology pair and using multiple pairwise PERMANOVAs. We also observed beta dispersal of our data as a further means of testing the assumptions of PERMANOVA analysis.

From the 126 *B. pascuorum* specimens, we obtained both gut-content and pollen sequence data for 23 individuals. Using this subset of paired samples, we compared the plant communities detected by the two metabarcoding methodologies at the individual sample level without aggregation. A PERMANOVA compared both methodologies' detected communities in strata defined by specimen of sample origin.

B. pascuorum-plant interaction network metrics

We used interaction frequencies from the three methodologies to build *B. pascuorum*-plant interaction networks and calculate species-level metrics for plant importance and specialization. Plant importance was defined as the proportion of all *B. pascuorum* interactions involving a given plant genus. For metabarcoding and pollen-load data, interactions were counted as the number of individual bee samples in which a plant genus was detected; for observational data, interactions corresponded to recorded visits. Species-level specialization (d') was calculated following Blüthgen et al. (2006), as implemented in

208 the R package bipartite (Dormann et al., 2009).

209 We created a composite interaction network for *B. pascuorum*, incorporating the data of each method-
210 ology and the interaction outcome types as network metadata. Network nodes included *B. pascuorum*
211 and the list of plant genera detected across the three interaction datasets. Single plant genera were
212 assigned between one and three links corresponding to interaction type, depending on their detection
213 across methodologies.

214 3. Results

215 *Assessment of floral resource use relative to availability*

216 Within our flower count surveys we registered a total of 117 flowering herbaceous plant genera across
217 the sampling season, representing the pool of floral resources available to *B. pascuorum*, which in-
218 teracted with only a subset of this diversity (Fig. 1). In fact, 39 genera recorded in flower counts
219 were absent from the interaction networks generated by any of the methodologies. Interaction transects
220 revealed interactions with 27 genera (23% of total floral diversity), while gut-content and corbicular
221 pollen metabarcoding revealed interactions with 58% and 53% of available taxa, respectively.

222 *Comparison of interaction detection by methodology*

223 Between all pollen metabarcoding samples, we obtained 5,160 ASVs within 2,506,925 reads, while in
224 122 of the gut-content metabarcoding samples, we identified 2,302 ASVs from 2,973,224 reads. Me-
225 dian sample coverage was 99% and 88%, for pollen and gut-content samples, respectively (See Sup-
226 porting Information). Before quality screening of results, 251 taxa were identified by both metabar-
227 coding methodologies in total. After screening, 170 total taxa remained identified within our data.

Both metabarcoding methodologies detected multiple unique taxa (33 taxa for gut contents and 30 for corbicular pollen), while interaction transects did not detect any unique interactions (Fig. 1). The two metabarcoding methodologies shared 83 common plant genera, representing 68% of the total corbicular pollen diversity and 63% of the gut content diversity.

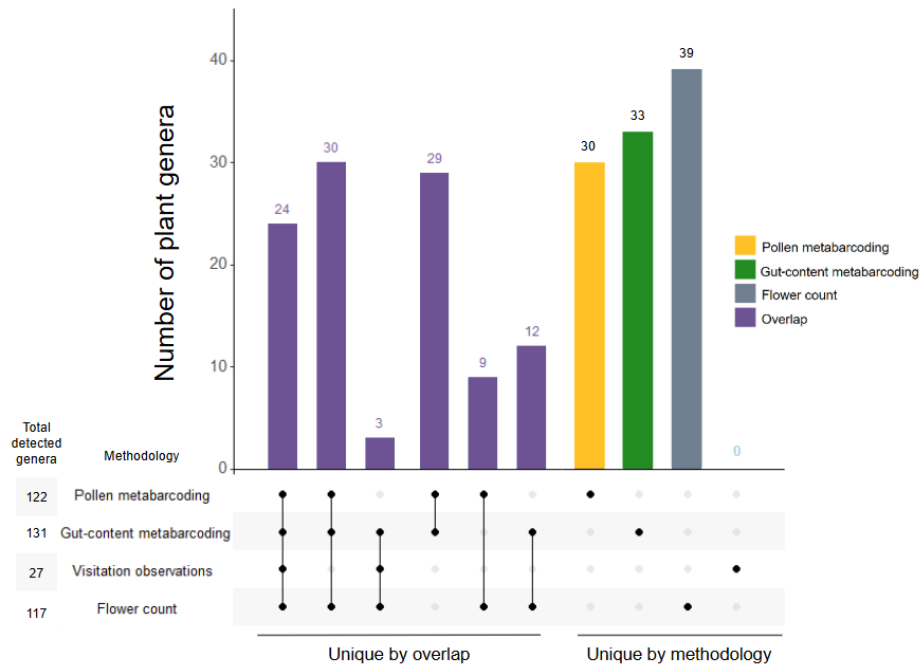


Figure 1: Total diversity and overlap of interaction plant genera observed by four observation methodologies: transect surveys of floral diversity (“Flower count”) and *Bombus pascuorum* - flower visitation, and metabarcoding of plant DNA in corbicular pollen and gut contents of *B. pascuorum*.

Taxonomic diversity varied across sampling periods, revealing distinct temporal patterns in flowering taxa and interactions (Fig. 2). Although floral and interaction diversity increased overall from the first to the last period, flowering taxa peaked in period four, whereas interactions peaked in period six. Metabarcoding consistently detected more taxa than interaction transects, with gut-content

metabarcoding outperforming all other methods. In periods one, two, and six—before and after peak
flowering—gut-content metabarcoding detected 59% more taxa than were recorded in flower counts,
while in periods three to five floral diversity exceeded gut-content diversity.

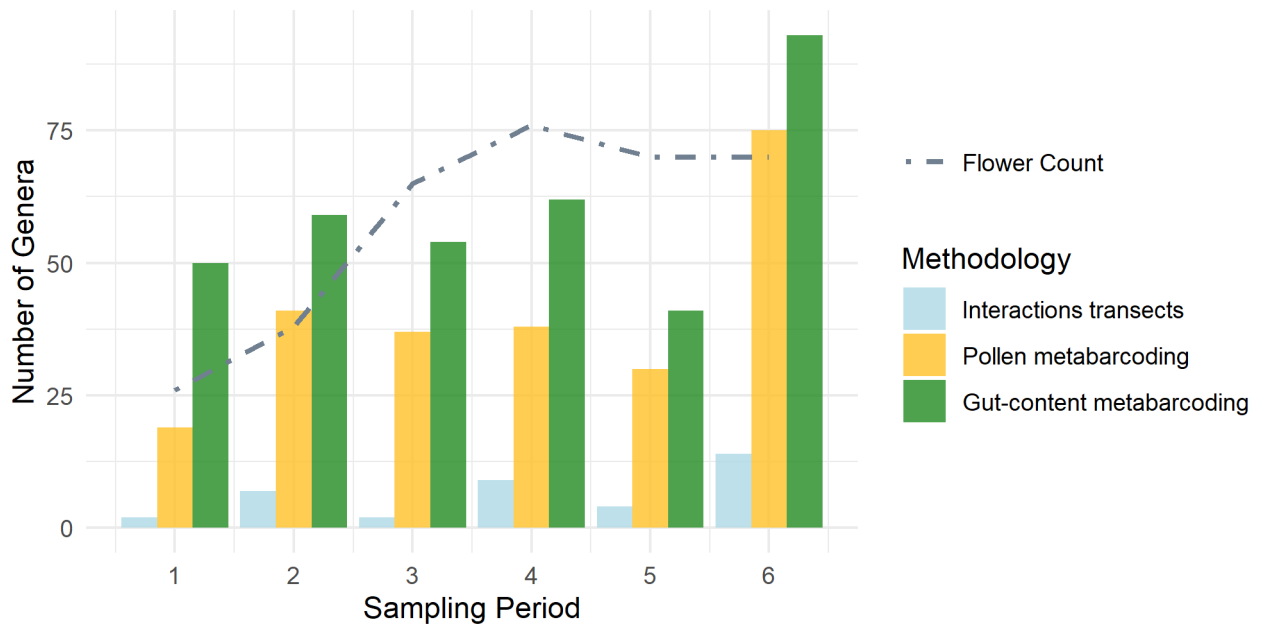


Figure 2: Taxonomic diversity in *Bombus pascuorum* interaction networks over six sampling periods (April - August, 2023) observed through floral visitation surveys and ITS2 metabarcoding of DNA extracted from bumblebee gut contents and corbicular pollen loads. The results of each methodology correspond to samples or surveys taken across the same 48 sampling days. The number of plant genera indicated is a cumulative raw value for each methodology and period, with no standardization for sampling effort. Interaction diversity for transects is represented by the total number of plant taxa observed over each transect and sampling day, for each period. For metabarcoding methodologies, interaction diversity is the total number of plant genera observed across all samples collected during the given period.

253 *Functional diversity observations*

254 The design of interaction transects only included taxa from the entomophilous community, while both
255 metabarcoding methodologies detected taxa from the anemophilous community as well, representing
256 28% (N = 41) of the total identified plant genera between the two methodologies. Among these were 20
257 genera from *Poaceae*, nine tree or woody plant genera, and 12 other herbaceous genera (See Supporting
258 Information Fig. S3). During periods one, two, and six—when gut-content metabarcoding detected
259 more taxa than the entomophilous community recorded in transects—an average of 13% of those taxa
260 were anemophilous or partially anemophilous (See Supporting Information, Fig. S4).

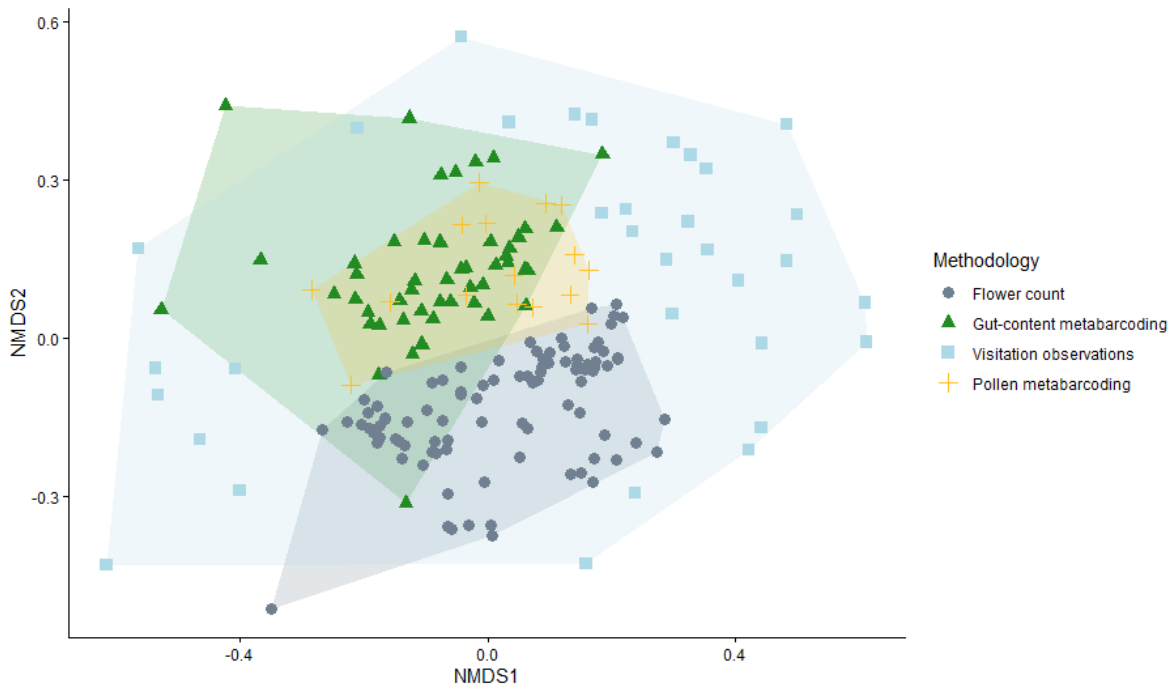
261 *Plant community composition across methodologies*

262 PERMANOVA indicated a significant effect of methodology on the observed community ($P < 0.001$, R
263 $= 0.28$). In this analysis, interaction transects showed high beta-dispersal (distance to centroid = 0.62)
264 compared to the more centered metabarcoding and flower count results (distance to centroid ≤ 0.10),
265 and an ANOVA test of mean dispersal by methodology indicated different levels of dispersal ($P < 0.001$)
266 for each methodology. The communities detected by each of the methodologies were also visualized
267 using non-metric Multidimensional Scaling (nMDS, stress = 0.17, Fig. 3). Pairwise comparisons (Table
268 1) showed that the plant communities detected by flower counts were different from those of all other
269 methodologies ($P < 0.001$, Holm-Bonferroni), although between pairs of interaction methodologies, no
270 differences were observed.

271 **Table 1.** *Pairwise tests comparing the community composition of plant taxa detected by four method-*
272 *ologies. Detected communities were compared by repeating PERMANOVA tests for each methodology*
273 *pair. Tests applied Raup-Crick dissimilarity to 9999 permutations, and adjusted p-values were calcu-*

lated using the Holm–Bonferroni method. The summarized test statistics include degrees of freedom
for each methodology (DF), R^2 , test F-statistics (F) and associated p-value (p), as well as the adjusted
p-value.

Methodology 1	Methodology 2	DF1	DF2	R^2	F	p	Adjusted p
flower count	gut metabarcoding	1	1	0.534	161.69	<0.001	<0.001
flower count	pollen metabarcoding	1	1	0.376	64.95	<0.001	<0.001
flower count	interaction	1	1	0.230	37.64	<0.001	<0.001
gut metabarcoding	pollen metabarcoding	1	1	0.130	9.38	0.0369	0.111
gut metabarcoding	interaction	1	1	0.010	0.80	0.55	1
pollen metabarcoding	interaction	1	1	-0.024	-1.13	0.997	1



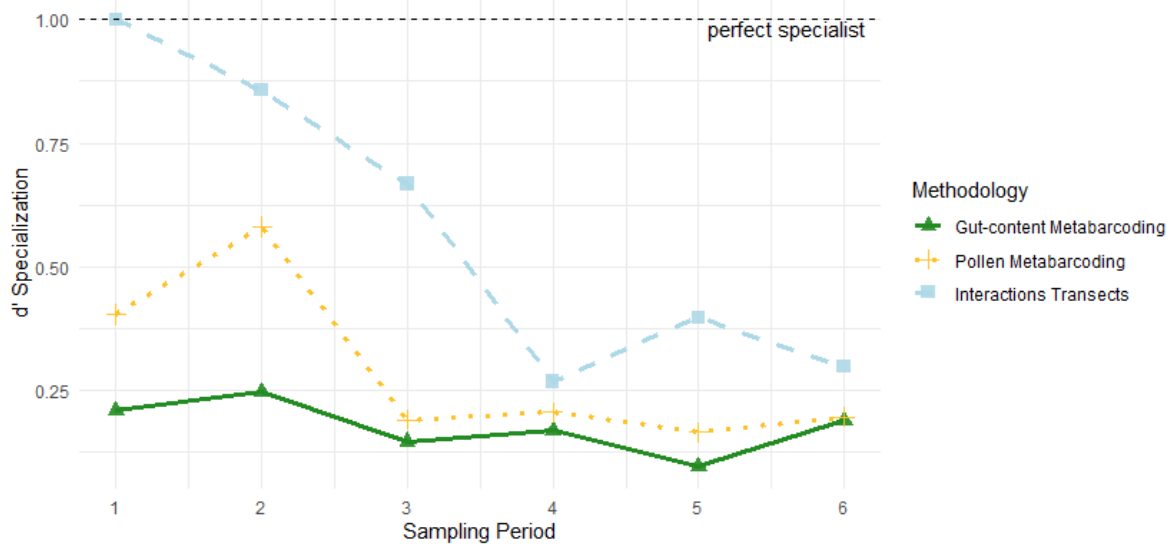
278 **Figure 3:** Non-metric multi-dimensional scaling plot of plant communities detected by three method-
279 ologies for observing *Bombus pascuorum* floral interactions and a flower diversity survey. Observa-
280 tions from each methodology are aggregated by sampling day, reduced to binary presence/absence data,
281 and compared in ordination using the Raup-Crick dissimilarity index (ordination stress = 0.17).

282 *Gut-content vs. pollen-derived plant communities from paired samples*

283 Comparing metabarcoding results from the same specimens, gut contents yielded fewer taxa (mean =
284 13 genera, sd = 9) than pollen samples (mean = 19 genera, sd = 7). On average, only 21% of taxa
285 (mean = 7 genera, sd = 3) were shared between the two sample types. A PERMANOVA with specimen
286 as a blocking factor indicated a difference in the plant community observed by both sample types (See
287 Supporting Information Table S2) explaining 16% of the variation between gut- and pollen-based de-
288 tections. Data used in this comparison were similarly dispersed (distance to centroid = 0.08), with no
289 difference between the two groups observed by a permutest.

290 *Species level interaction network*

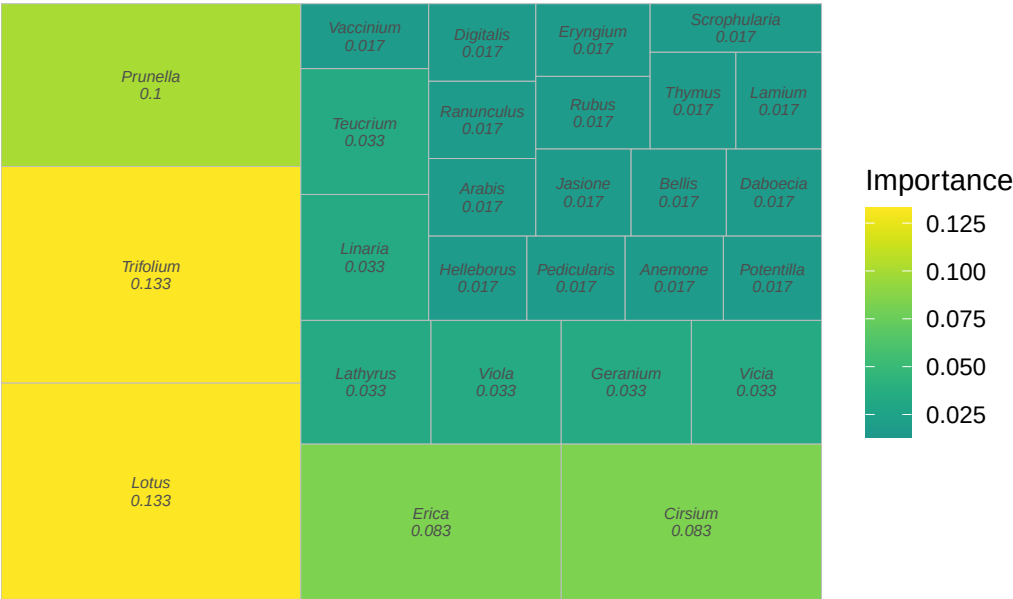
291 We calculated interaction specialization of *B. pascuorum* and an importance metric for the plant taxa
292 within interaction networks. Specialization [d' ; Blüthgen et al. (2006)] declined over the season for
293 transect and pollen-metabarcoding data but remained relatively stable for gut-content metabarcoding
294 (Fig. 4), with transects indicating complete specialization in the first period. Across all methods, *Lo-*
295 *tus* emerged as the most important plant genus, though the structure of importance differed: the two
296 metabarcoding networks showed more evenly distributed importance values (Fig. 5b-c), whereas the
297 transect network was dominated by a few top taxa (Fig. 5a).



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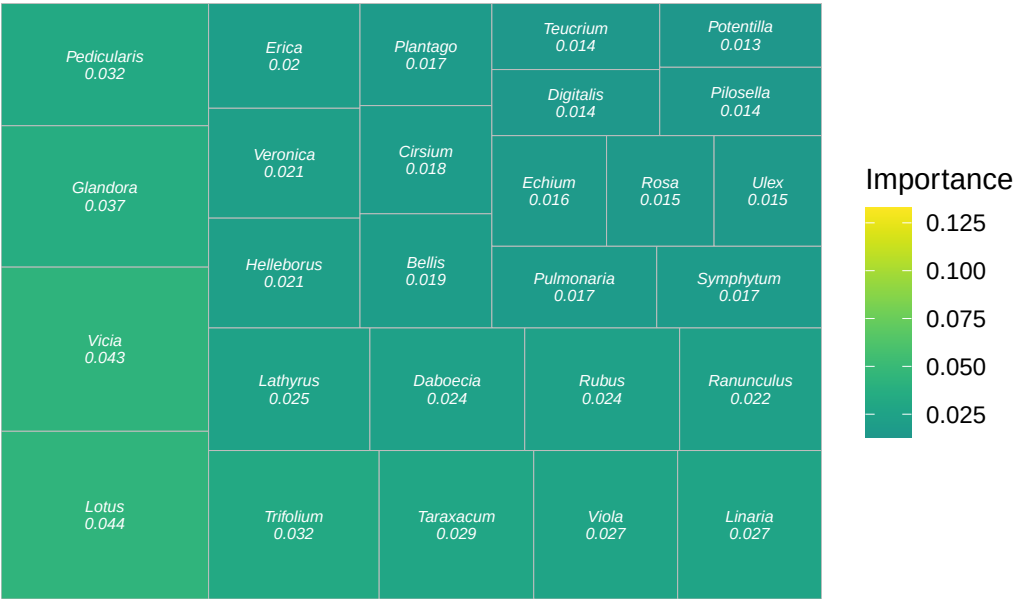
299 **Figure 4:** Specialization of plant interactions for *Bombus pascuorum* as indicated by networks con-
 300 structed from three interaction observation methodologies. Specialization was calculated as d' using
 301 the methodology of Blüthgen et al. (2006), with $d' = 1$ representing perfect specialist behavior. Spe-
 302 cialization of *B. pascuorum* for each period was calculated relative to interaction data from the same
 303 species in other periods, rather than other pollinator species.

A. Interaction transect-based network



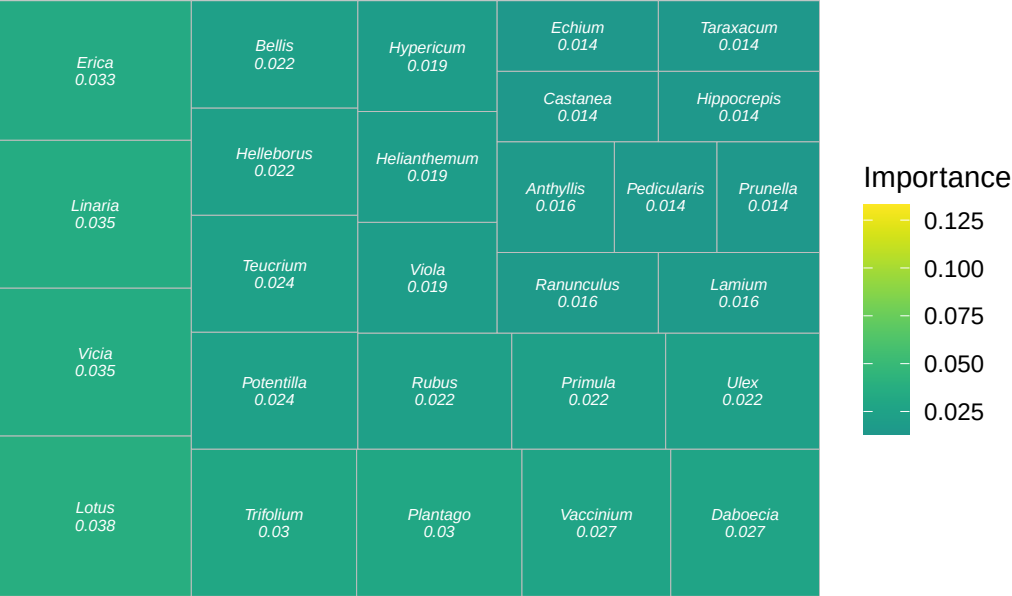
304

B. Gut-content metabarcoding



305

C. Corbicular pollen metabarcoding



306

307 **Figure 5.** Plant “importance” within *Bombus pascuorum* interaction networks constructed from (A)
308 interaction transects, (B) gut-content metabarcoding, and (C) corbicular pollen metabarcoding. Im-
309 portance was calculated as the proportion of total plant interactions observed by the given method-
310 ology represented by interactions with the specific plant genus. Importance is visualized with block
311 size proportional to importance, and color scaled to minimum and maximum values observed by each
312 methodology.

313 Combined interaction network

314 We combined the results from each interaction methodology to create an interaction network for *B.*
315 *pascuorum* with links defined by interaction outcomes, including consumption, transport, and visitation
316 (Fig. 6). This single species network included 171 nodes, increasing the number of taxa included in
317 the network compared to individual methodology constructed networks. Additionally, each plant taxa

received up to three links, including link metadata for interaction outcomes in the network. In total, the network contained 281 descriptive links.

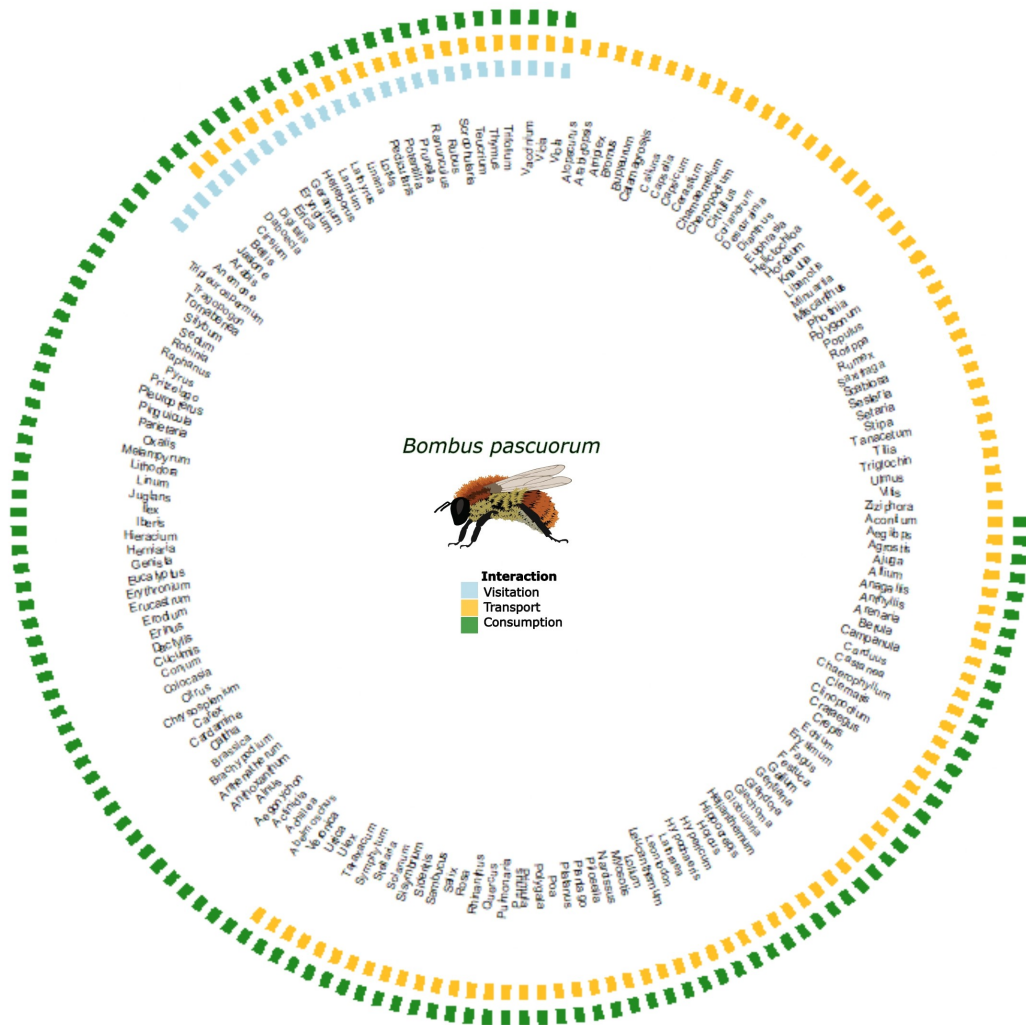


Figure 6. Combined interaction network for *Bombus pascuorum* including all interaction plant taxa detected by three methodologies. Interaction transect observations are represented by visitation, corbicular pollen metabarcoding observation by transport, and gut-content metabarcoding observations

324 *by consumption. The network includes 170 plant genera, each with up to three links describing the*
325 *outcomes of interactions. Outcome links represent the presence of any interaction observation within*
326 *the dataset of the corresponding methodology.*

327 **4. Discussion**

328 Plant-pollinator interaction networks are complex. We show that using functionally informa-
329 tive network links and combining methodologies yields a comprehensive and strongly validated
330 plant-pollinator interaction network. Notably, the two metabarcoding approaches revealed shared in-
331 teractions with anemophilous and partially anemophilous plants for pollen consumption and transport,
332 highlighting the complementarity of their data. Although each interaction methodology overlapped
333 statistically at the aggregated level, the combined network resulting from each methodology increased
334 the total nodes, and each methodology provided context to network links. Metabarcoding alone
335 also proved effective at capturing a broad range of links and providing detailed, specimen-level
336 data. Important information from the function of links is missing under the current approach to
337 characterizing interaction networks, but using multiple methodologies helps to fill these gaps.

338 We compared each methodology in terms of the diversity of detected interactions, assignment of relative
339 importance of plant taxa and specialization of *B. pascuorum* within the resulting network, and observed
340 plant community composition. Consistent with previous comparisons between field and metabarcod-
341 ing observation of plant-pollinator interactions, metabarcoding increased observed interaction diversity
342 (Baksay et al., 2022; Milla et al., 2022; Smart et al., 2017), in our case by more than six-fold compared
343 to interaction transect results. Considering this, and the time dedicated to data collection for both types
344 of methodologies, metabarcoding was a more efficient approach. Interaction transects did provide the

345 advantage of greater taxonomic resolution, as we were able to detect interactions at the species-species
346 level, whereas metabarcoding provided species-genus level interactions. Beyond taxonomic detection
347 capabilities, the results from each methodology allowed for network-level cross-validation.

348 Network topology and specialization patterns differed markedly across methodologies. Interaction tran-
349 sects tended to overstate both the degree of specialization and the dominance of the most frequently
350 visited plant taxa. Although *B. pascuorum* is known to form strong early-season associations with cer-
351 tain plant species (Artamendi et al., in prep), the metabarcoding approaches indicated much lower spe-
352 cialization and produced more evenly distributed network structures. These results mirrored previous
353 interaction networks constructed for individual pollinator species, which also have shown a tendency
354 towards representing pollinators as specialists when using field observation data versus the generalist
355 behavior indicated by metabarcoding data (Arstingstall et al., 2021). Overall, the combined datasets
356 across methodologies suggested a more diverse foraging niche than visitation data alone would have
357 implied.

358 The three methodologies showed complementary patterns in network composition. Flower counts and
359 interaction transects overlapped as expected from the study design, yet differed statistically, likely due
360 to the much larger number of taxa detected by the former. No statistical differences were found among
361 the three interaction-focused methods, although their dispersion differed, reflecting variation in spa-
362 tial and taxonomic coverage. Interaction transects are shaped by local habitat and plant-community
363 differences, whereas metabarcoding integrates interactions across the broader landscape, producing
364 more consistent results. Metabarcoding approaches overlapped minimally with the floral community
365 detected by flower counts, indicating that interaction networks include taxa not captured within tran-

sects. This is unsurprising given that flower counts reflect potential, not actual, interactions and are constrained by spatial and temporal limits that do not restrict metabarcoding.

Between the two metabarcoding approaches, gut-content metabarcoding captured greater overall taxonomic diversity and was more efficient, given that every specimen provided a gut sample, but not necessarily a pollen sample. Pollen samples detected more taxa per individual, however, and hypothetically offered an advantage as a non-lethal sampling option. The combination of both methodologies' results broadened the interaction network greatly, and incorporated contextualized interaction links. These links showed which plant genera were consumed for adult bee nutrition, and which provided pollen for transport to the nest. In our case, gut-content metabarcoding was particularly informative for revealing seasonal foraging patterns, detecting more consumed taxa than were flowering in the early and late parts of the season, and showing relatively stable low specialization over time. This stability contradicted previous field observations, indicating higher specialization of the species, especially in the early flowering season (Artamendi et al., unpublished data). Together, these results indicated that the plant community represented in consumption-based interactions differs from the floral community captured by field- and pollen-based surveys.

Metabarcoding observes forage across functional groups

The diversity of plant groups observed within our metabarcoding data, especially the temporal changes in diversity observed by gut-content metabarcoding, indicated that *B. pascuorum* forages on different plant taxa than previously expected. Through metabarcoding, we observed interactions with a variety of taxa outside of the entomophilous meadow and shrubland plant community, including trees and shrubs, grasses, and other herbaceous plants. Our observations of anemophilous plant interactions are

387 supported by previously documented records for *Bombus* species (de Vere et al., 2017; Ibiyemi et al.,
388 2025; Milla et al., 2022; Pojar, 1973; Selva et al., 2024; Tanaka et al., 2020; Terrell & Batra, 1984;
389 Timberlake, de Vere, et al., 2024; Wood et al., 2022), and have especially intriguing implications for
390 bumblebee forage behavior. Previous studies using external pollen metabarcoding have removed wind-
391 pollinated taxa from their analyses under the argument that wind-borne pollen in samples may represent
392 false positive interactions (Negri et al., 2015; Pornon et al., 2017; Tanaka et al., 2020). Our gut-content
393 results, however, caution against the practice of removing these taxa as contaminants, especially if using
394 external pollen loads as standalone proxies for forage networks.

395 The presence of DNA from anemophilous taxa within gut samples suggests that interactions with these
396 taxa may be more than coincidental interactions with pollen in the environment. Indeed, beyond con-
397 sumption for adult nutrition, there are previous indications that pollen from flowering trees supports
398 colony establishment success and low larval mortality (Wood et al., 2022). Our results support the hy-
399 pothesis that most bumblebees forage selectively for consumption and transport of high quality pollen
400 (Ruedenauer et al., 2016; Timberlake, de Vere, et al., 2024), adapting their forage to take advantage of
401 the best available resources as they change with environmental variability (Selva et al., 2024). While
402 it is possible that some plant material may be transported or consumed incidentally (Arstingstall et al.,
403 2021), the taxa detected within *B. pascuorum* gut contents and corbicular pollen form part of the web
404 of biodiversity that supports the species and possibly other pollinators. Our detection of anemophilous
405 plant DNA in both metabarcoding methodologies indicates that *B. pascuorum*, and perhaps other bum-
406 blebee species, may intentionally forage on these taxa to meet nutritional needs at various life stages.

407 Existing hypotheses for pollinator forage adaptations in response to environmental changes have sug-

gested that bees expand forage diversity beyond the flowering community and across habitats in order to survive annual “hunger gaps” (Becher et al., 2024; Timberlake, Tew, et al., 2024), when blooming floral species are limited (Morozumi et al., 2022; Wood et al., 2022). Our observation of high forage diversity in gut contents before and after the floral peak, distinct interaction and flowering taxa network topologies, and consumption of taxa across functional groups, all together support these hypotheses. While the floral community beyond the immediate area of our transects likely played a large role in these observations, the detection of anemophilous taxa in gut contents during the periods where forage diversity was higher than flowering diversity provide evidence for a community driven component as well. These observations show how the broader taxonomic detection capacity of metabarcoding allows for detection of interactions that otherwise would go unobserved by flower visitation surveys. This advantage is extended when working with metabarcoding data at the individual sample level, where greater resolution for interactions is obtainable.

Metabarcoding offers individual level analysis

Our comparative analyses understate the resolution of the metabarcoding derived data. We aggregated detections by sampling day to balance effort across methods, overlooking the individual-level detail that metabarcoding can provide. When we compared taxa detected from paired pollen and gut samples at the individual level, overlap was low, revealing a difference between sample sources that was not apparent in comparisons of aggregated data. This difference likely reflects the different roles of corbicular pollen and immediately consumed pollen in the nutrition needed for different life-cycle stages (Vaudo, 2015). Taxa repeatedly detected by both methods increased confidence in their importance. For instance, the consistent appearance of *Vicia* in both sample types early in the season supports field observations of a

strong association between *B. pascuorum* and *Vicia* species (Artamendi et al., unpublished data), again underscoring the value of integrating field surveys with laboratory-based methods.

Metabarcoding also has important limitations. Primer and marker biases result in preferential amplification of certain sequences, making some taxa more or less likely to appear in final results. We used the ITS2 region as a target for amplicon sequencing, while other marker genes provided competitive options for targeting plant taxa (Espinosa Prieto et al., 2024). Finally our reference database inevitably contains gaps in coverage of available sequences for matching our data to ASVs and taxa. In response to these limitations, we further suggest that combination with other interaction methodologies is likely to be the most effective way to overcome inherent methodological oversights.

Conclusions

The similarities between interaction data suggest robustness between each methodology, and the inherent implications of the sample sources of each provide varied means of interpreting different interactions. Interaction transects provide a valuable field-based perspective, although given their lower sampling efficiency, incorporating them as a validation of other surveys may be the best way to integrate this methodology into future studies. Field observations can fill gaps left by metabarcoding methodologies, such as confirmation of pollination efficacy, interaction frequency, and species-level resolution. As a direct observation of the pollen transported to the nest, corbicular pollen may also be a good starting point for identifying plants that may provide pollen with optimal macronutrients for larval development. Similarly, gut-content metabarcoding provides an important perspective on the nutritional needs of actively foraging pollinators, identifying taxa that provide pollen as food for supporting this activity (Li et al., 2025). Knowing which taxa are actually ingested by pollinators is

450 especially useful for identifying taxa that facilitate microbiota exchange and acquisition during plant
451 interactions (Cullen et al., 2021; Keller et al., 2021), including parasite and disease transfer (Lignon et
452 al., 2024). Although they are not equal, our research highlights overall that each methodology offers
453 advantages and disadvantages in terms of sensitivity, sampling effort, and perspective.

454 While most of the methodologies we applied, aside from gut-content metabarcoding, have previously
455 been used independently to characterize plant–pollinator networks (e.g., Devriese et al., 2024; Magrach
456 et al., 2023), our findings highlight the synergistic value of integrating them. Gut-content metabarcod-
457 ing emerges as a promising approach, but its greatest potential is realized when combined with estab-
458 lished approaches. A key next step is improving our ability to quantify interaction frequencies at the
459 individual level using metabarcoding, whether from gut contents or pollen. Overall, methodological
460 advances are likely to come from linking complementary data sources to fill the informational gaps left
461 by any single approach.

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